

ZERO TRUST ARCHITECTURE (ZTA) QA GAP ANALYSIS

Important Scope Disclaimer: All testing and evaluations referenced in this analysis were executed strictly within my own custom, in-house lab environment. No tests were conducted using an official or live Instasafe demo portal.

1. What I Have Tested So Far

In my custom lab setup, the test coverage focused heavily on initial authentication and network lock-down:

Secure Tunnels: I verified that all network communication is encrypted using mTLS and that unauthorized traffic is dropped by default.

Identity & Login: I tested multi-factor authentication (MFA) and basic rules like blocking logins based on time or location.

Static Health Checks: I ran simple pre-login checks to confirm an endpoint's OS and antivirus state before granting access.

2. Key Testing Gaps (What Was Missed)

Because my tests focused primarily on the initial login step, two major areas remain undertested:

Continuous Device Health Monitoring: I verified device safety at login, but I did not test what happens if a device gets infected or turns off its security software while connected.

Automated Risk Response: I collected event logs, but I never tested whether suspicious activity automatically triggers the system to block a user or kill a session.

3. The Risk

The primary rule of Zero Trust is "never trust, always verify." By only testing initial login, I leave a major blind spot. If an attacker compromises a laptop after a legitimate user logs in, my current test cases wouldn't catch it. The active session would remain open, allowing unauthorized access.

Recommended Test Scenario

Title: Mid-Session Disconnect on Antivirus Disablement

Objective: Start an active file transfer inside my custom lab environment, then manually turn off the antivirus service on the test laptop. Verify that the gateway detects the missing security health signal and automatically drops the connection within 30 seconds.